

CS3 Code Review - Lab 4 Roster

Name: _____

Date: _____

Choose the best answer for each question. Assume the Lab 4 roster programs are used exactly as provided.

1. If the program is invoked as shown below, what is the value of argc?

```
classSchedule cs1.txt cs2.txt dropouts.txt
```

- A. 3
- B. 4
- C. 5
- D. It depends on the number of students in the files

2. In the procedural version, what does one StudentEntry contain?

- A. A course name followed by all students taking that course
- B. A student's full name followed by that student's course names
- C. Only a student's first and last name
- D. A filename followed by its file extension

3. What does `courseName("C:\rosters\cs2.txt")` **return?**

- A. C:\rosters\cs2
- B. cs2.txt
- C. cs2
- D. .txt

4. A student named Ava Reed appears in both cs1.txt and cs3.txt. What will the procedural program store for her?

- A. Two separate entries, one for each course
- B. One entry containing Ava Reed, cs1, and cs3
- C. One entry containing only the last course, cs3
- D. No entry, because duplicate students are removed entirely

5. Why does `addCourse()` **call** `std::find()` **before** `courses_.push_back(course)`?

- A. To keep the courses in alphabetical order
- B. To prevent a repeated name in one input file from adding the same course twice
- C. To locate the student in allStudents
- D. To remove a course when a student drops out

6. In the object-oriented version, what happens when this statement executes for a new student?

```
allStudents.push_back(std::move(student));
```

- A. The Student move constructor transfers the strings and course list into the new list element
- B. The student is copied, and the source is guaranteed to remain unchanged
- C. The source object is destroyed before insertion begins
- D. Only the student's first name is inserted

7. Why does `operator==` compare only `firstName_` and `lastName_`, not `courses_`?

- A. Course lists cannot be compared in C++
- B. Students with the same name must match when merging rosters or removing dropouts
- C. Comparing courses would automatically sort them
- D. `std::find` ignores the equality operator's result

8. According to the object-oriented `operator<`, which student appears first after `allStudents.sort()`?

- A. Zoe Adams
- B. Amy Young
- C. They compare equal because only last names are checked
- D. Their order is unpredictable

9. After `allStudents.splice(allStudents.end(), lst)` executes in `roster.cpp`, what is true?

- A. The elements are copied into `allStudents`, and `lst` keeps its elements
- B. The elements are transferred into `allStudents`, leaving `lst` empty
- C. Both lists share ownership of the same nodes
- D. `allStudents` is automatically sorted and deduplicated

10. Why must `allStudents.sort()` occur before `allStudents.unique()` in `roster.cpp`?

- A. `unique()` removes only adjacent equal elements, so sorting brings duplicates together
- B. `unique()` works only with vectors that have first been sorted
- C. `sort()` deletes duplicates, while `unique()` restores one copy
- D. The compiler rejects a call to `unique()` on an unsorted list

Answer Key

1	B
2	B
3	C
4	B
5	B
6	A
7	B
8	B
9	B
10	A